Relational Model and Relational Algebra

Relational Model

- The **relational model** was first introduced by **Ted Codd of IBM in1970**
- The model uses the concepts of a mathematical relation (table of values) as its basic building block.
- The model has its theoretical basis
  - set theory
  - first –order predicate logic.

Relational Model

•The model has been implemented in a large number of commercial DBMS.

1. Oracle DBMS and SQL /DS DBMS (by IBM)
2. DB2 and Informix Dynamic Server (IBM)
3. Oracle and Rdb (Oracle)
4. SQL Server and Access(Microsoft)

- **Our discussion is divided into 2 parts :**
  - Relational model concepts
  - Operations on relational model using relational algebra

Topics to be discussed

1. **Relational Model Concepts**
2. **Relational Model Constraints and Relational Database Schema**
3. **Update operations and Dealing with Constraint Violations**
4. **Unary Relational Operations (SELECT and PROJECT)**
5. **Relational operations from Set theory**
6. **Binary Relational Operations**
7. **Examples of Queries in Relational operations.**
8. **Relational Database Using ER-to -Relational Mapping**

Relational Model Concepts

Relational model terminology

- The relational model represents the database as a collection of relations.
  - a row is called a ***tuple***,
  - a column header is called an ***attribute***
  - the table is called a ***relation.***
  - The data type describing the types of values that can appear in each column is called a ***domain***

Relational model terminology

Relation name

Attributes

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

Tuples

Domains

- A domain D is a set of atomic values. A domain has name, data type and format.

Examples for domains

1. **Phone_numbers :** A set of 10 digit numbers
2. **Social_securiry_numbers:** The set of valid nine-digit social security numbers
3. **Employee_ages:** set of integers between 16 and 60
4. **Academic_Department_names**: The set of academic department names in a university, such as Computer Science, Economics, and Physics.

Domains

- A domain has a **name** ,a **data type and a format**

Examples : Phone_numbers

Data type : a character string

format : dddd-dddddd

where each d is a digit

Employee_ages (Data type : Integer)

Academic_Department_names

data type : a set of Character String

Relation Schema and Attributes

- A **relation schema** R, denoted by **R(A1, A2, . . ., An),** is made up of a relation name R and a list of attributes A1, A2, . . ., An.

- Each **attribute** A_i is the name of a role played by some domain D in the relation schema R. D is called the **domain** of Ai and is denoted by **dom(A_i)**.

- The **degree (arity)** of a relation is the number of attributes (n) of its relation schema.

An example : University students

STUDENT(Name, SSN, HomePhone, Address, OfficePhone, Age, GPA)

Relation Name : STUDENT

Attributes : Name, SSN, HomePhone, Address, OfficePhone, Age, GPA

Domains:

- dom(Name) = set of all Names;
- dom(SSN) = Social_security_numbers;
- dom(HomePhone) = Local_phone_numbers,
- dom(HomePhone) = Addresses
- dom(OfficePhone) = Local_phone_numbers,
- dom(GPA) = Grade_point_averages.

Degree of the relation : 7

Relation state

- **A relation state r** of the relation schema R(A1, A2, . . ., An), denoted by **r(R),** is a set of n-tuples r = {t1, t2, . . ., tm}.
- Each **n-tuple** t is an ordered list of n values t = <v1, v2, . . ., vn>, where each value v_i, 1<=i<= n, is an element of dom(A_i) or is a special null value.

A relation state r of student relation schema R

Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

Relation state

- The i^{th} value in tuple t, which corresponds to the attribute A_i, is referred to as $t[A_i]$.
- The terms **relation intension** for the schema R and **relation extension** for a relation state r(R) are also commonly used.

More Formally a relation state can be restated as follows:

A relation (relation state) r(R) is a mathematical relation of degree *n* on the domains dom(A1) ,dom(A2) …dom(An) , which is a subset of Cartesian Product of the domains that define R.

$$r(R) \subseteq (\mathrm{dom}(A_1) \times \mathrm{dom}(A_2) \times \ldots \times \mathrm{dom}(A_n))$$

Characteristics of Relations

1. Ordering of Tuples in a Relation

- A relation is a set of tuples. Hence , tuples in a relation do not have any particular order.

- Tuple ordering is not part of a relation definition. But many logical ordering can be specified on a relation.

- When a relation is implemented as a file or displayed as table , a particular ordering may be specified on the records of the file or on the rows of a table.

3.2 Characteristics of Relations

1. Ordering of Tuples in a Relation

Relation name

Attributes

Tuples

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21

The above relations are identical (i.e tuple ordering does not matter)

Characteristics of Relations

2. Ordering of Values within a Tuple

- An n-tuple is an *ordered list* of n values. Hence the ordering of values of an n-tuple is important.

- However at a logical level ,the order of attribute and their values is not important as long as the correspondence between the attributes and values is maintained.

Characteristics of Relations

The order of attribute and their values is not important as long as the correspondence between the attributes and values is maintained.

The following two tuples are identical

t = < (Name, Dick Davidson),(SSN, 422-11-2320),(HomePhone, null),(Address, 3452 Elgin Road),
(OfficePhone, 749-1253),(Age, 25),(GPA, 3.53)>

t = < (Address, 3452 Elgin Road),(Name, Dick Davidson),(SSN, 422-11-2320),(Age, 25),
(OfficePhone,749-1253),(GPA, 3.53),(HomePhone, null)>

Characteristics of Relations

3. Values and nulls in the Tuples

- In relation each value in a tuple is an **atomic** value; (indivisible)
- Hence composite and multi valued attributes are not allowed. That is why this model is called **flat relational model.**
- Multivalued attributes must be represented in a separate relation.
- Composite attributes are represented only by their simple component. (e.g. fname)

Characteristics of Relations

3. Values and Nulls in the Tuples

- The values of some attributes within a particular tuple may be unknown or may not apply to that tuple. A special value, called **null** can be used in this case.

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21

- Some student do not have office , so officephone number is null (not applicable)
- Some students do not have homephone number or have one but we do not know, so homephone number is null (unknown),

Characteristics of Relations

4. Interpretation (Meaning) of a Relation :

a) **The relation schema can be interpreted as a *declaration or a type of assertion*. Each tuple in the relation can then be interpreted as a *fact or a particular instance of the assertion*.**

Relation name

Attributes

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

Tuples

Characteristics of Relations

4. .

5. Interpretation (Meaning) of a Relation :

- **Relations may represent facts about entities, whereas other relations may represent facts about relationships .**

  For example: A relation schema `MAJORS (StudentSSN, DepartmentCode)` asserts that students major in academic departments; a tuple in this relation relates a student to his or her major department.

b. Alternative interpretation of a relation schema is as a **predicate.** In this case values in each tuple are interpreted as values that satisfy the predicate

Relational Model Notation :

- The letters Q, R, S denote relation names.
- The letters q, r, s denote relation states.
- The letters t, u, v denote tuples.
- A relation schema R of degree n is denoted by R(A1, A2, . . ., An).

Where A1,A2...An are the attributes of R

Relational Model Notation :

- An **n-tuple** t in a relation r(R) is denoted by **t = <v1, v2, . . ., vn>** where v_i is the value corresponding to attribute A_i. The following notation refers to component values of tuples:

  - Both $t[A_i]$ and $t.A_i$ refer to the value v_i in t for attribute A_i.
  - Both t[Au, Aw, . . ., Az] and t.(Au, Aw, . . ., Az), where Au, Aw, . . ., Az is a list of attributes from R, refer to the subtuple of values $<v_u, v_w, \ldots, v_z>$ from t corresponding to the attributes specified in the list.

Relational Model Notation

- An attribute A can be qualified with the relation name R to which it belongs by using the dot notation R.A

for example : STUDENT.Name or STUDENT.Age.

This is because the same name may be used for two attributes in different relations.

Relational Model Notation : Example

Relation Schema :

STUDENT (name,SSN,HoemPhone,Address,OfficePhone,Age,GPA)

Relation State r(STUDENT)

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

t4 **=** < Charles Cooper,489-22-1100,376-9821,265 Lark Lane , 749-6492,28,3.92>

t4[SSN] or t4.SSN = 489-22-1100
t3[Age] or t3.Age = 25
t1[Name] or t1.Name = Benjamin Bayer

Relational Model Constraints and Relational database schemas.

Relational Model Constraints

- Constraints generally specify the restriction on the actual values in a database.
- Constraints on databases can generally be divided into 3 main categories

1. Inherent Model based constraints
2. Schema - based constraints (we discuss only this)
3. Application - based constraints

Data dependency constraints :

1. Functional dependencies
2. Multi valued dependencies

Relational Model Constraints

1. Inherent Model based constraints :Constraints that are inherent in the data model

2. Schema - based constraints : Constraints that can be directly expressed in the schemas of the data model

   → Includes :

   1. Domain Constraints
   2. Key Constraints and Constraints on Null Values
   3. Entity Integrity, Referential Integrity constraints

3. Application-based constraints : Constraints that cannot be directly expressed in the schemas of the data model

1. Domain constraints :

- Domain constraints specify that within each tuple ,the value of each attribute A must be ***an atomic value*** from the domain dom(A).

- **The data types associated with domains include :**

1. standard numeric data types for integers (short-integer, integer, long-integer)
2. real numbers (float and double-precision float).
3. Characters, fixed-length strings, and variable-length strings.
4. date, time, timestamp, and sometimes money data types.
5. enumerated data type

2. Key Constraints and Constraints on Null

Superkey: A superkey SK specifies a uniqueness constraint that no two distinct tuples in any state r of R can have the same combination of values for all the attributes in SK.

Formally,

Consider **R {A,B,C,D}**
Let **r(R)** be a relation state of **R**
Let **SK** denote any one of the subsets of **R** , say **SK = {A,C}.**

Then **SK** is Super Key of **R** if, for any distinct tuples **t1** and **t2** in **r** ,

$$\mathbf{t1[SK] \neq t2[SK].} \quad (\mathbf{t1[A,C] \neq t2[A,C]})$$

*Every relation has at least one default superkey -**the set of all its attributes**.*

2. Key Constraints and Constraints on Null

Superkey: Example

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

The above is a relation state of STUEDNT relation. i.e. r(STUDENT)

The subset SK = {SSN, name , Age) , is a superkey of STUDENT, because for ***any two distinct*** tuples t1,t2, in r(STUDENT) ,

t1[SK] ≠ t2[SK].

2. Key Constraints and Constraints on Null

Key **:** A **key K** of a relation schema **R** is a superkey of **R** with the additional property that removing ***any attribute*** **A** from **K** leaves a set of attributes **K'** that is not a superkey of R.

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

{SSN} is a key of STUDENT (also a Super Key of STUDENT)

but {SSN , name ,age} not a key . Why?

2. Key Constraints and Constraints on Null

Hence, a key satisfies two constraints:

1. Two distinct tuples in any state of the relation cannot have identical values for (all) the attributes in the key.

2. It is a minimal superkey-that is, a superkey from which we cannot remove any attributes and still have the uniqueness constraint in condition 1 hold.

A key is determined from the ***meaning of the attributes****, and the property is* ***time-invariant.***

2. Key Constraints and Constraints on Null

supper key examples

CAR	LicenseNumber	EngineSerialNumber	Make	Model	Year
	Texas ABC-739	A69352	Ford	Mustang	96
	Florida TVP-347	B43696	Oldsmobile	Cutlass	99
	New York MPO-22	X83554	Oldsmobile	Delta	95
	California 432-TFY	C43742	Mercedes	190-D	93
	California RSK-629	Y82935	Toyota	Camry	98
	Texas RSK-629	U028365	Jaguar	XJS	98

SuperKeys of CAR:

1. { LN, ESN, Make, Model, Year} (not a key)
2. {LN, Year} (Not a key)
3. {ESN, Model, Year} (Not a key)
4. {ESN} (Also a Key)

Is Year a Super key of CAR?

2. Key Constraints and Constraints on Null

Key examples

Keys of Car relation :

Key1 = {LN}

Key2 = {ESN}, which are also superkeys of CAR.

BUT {LN, Make} is *not* a key , but is a superkey

- A relation schema may have more than one key. In this case, each of the keys is called a **candidate key** (LN and ESN are candidate keys)**.**

- If a relation has several **candidate keys**, one is chosen arbitrarily to be the **primary key**. The primary key attributes ***underlined*.**

2. Key Constraints and Constraints on Null

Constraints on Null :

Specifies whether null values are or are not permitted.

For example, if every STUDENT tuple must have a valid, non-null value for the Name attribute, then Name of STUDENT is constrained to be NOT NULL.

STUDENT	Name	SSN	HomePhone	Address	OfficePhone	Age	GPA
	Benjamin Bayer	305-61-2435	373-1616	2918 Bluebonnet Lane	null	19	3.21
	Katherine Ashly	381-62-1245	375-4409	125 Kirby Road	null	18	2.89
	Dick Davidson	422-11-2320	null	3452 Elgin Road	749-1253	25	3.53
	Charles Cooper	489-22-1100	376-9821	265 Lark Lane	749-6492	28	3.93
	Barbara Benson	533-69-1238	839-8461	7384 Fontana Lane	null	19	3.25

Relational Databases and Relational Database Schemas

Relational Database Schema : A relational database schema S is a set of relation schemas S = {R1, R2, . . ., Rm} and a set of integrity constraints IC.

EMPLOYEE

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO

DEPARTMENT

DNAME	DNUMBER	MGRSSN	MGRSTARTDATE

DEPT_LOCATIONS

DNUMBER	DLOCATION

PROJECT

PNAME	PNUMBER	PLOCATION	DNUM

WORKS_ON

ESSN	PNO	HOURS

DEPENDENT

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP

Schema diagram for the COMPANY relational database schema

Relational Databases and Relational Database Schemas

Relational database state : A relational database state, DB, of S is a set of relation states DB = $\{r_1, r_2, \ldots, r_m\}$ such that each r_i is a state of R_i satisfies the integrity constraints specified in IC.

Invalid database state : A database state that does not obey all the integrity constraints is called an invalid state,

Valid database state : A state that satisfies all the constraints in IC is called a valid state.

Relational database state : example

One possible database state for the COMPANY database schema

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Relational Databases and Relational Database Schemas

Note the following :

1. Attributes that represent the same real-world concept may or may not have identical names in different relations.

   E.g. DNO in EMPLOYEE and DNUM in PROJECT .

2. Attributes that represent different concepts may have the same name in different relations.

   E.g. use NAME for PNAME of PROJECT and DNAME of DEPARTNAME.

3. If there are identical attribute names in real-world concept, then these attributes must have different names.

3. Entity Integrity, Referential Integrity

Entity integrity : The entity integrity constraint states that no primary key value can be null.

- *Key and entity integrity constraints are specified on individual relations.*

Referential integrity : The referential integrity constraint states that a tuple in one relation that refers to another relation must refer to an existing tuple in that relation.

- *Referential integrity is specified between two relations*
- *It is used to maintain the consistency among the tuples in two relations*

Referential Integrity can be defines more formally using Foreign Keys

Foreign keys :A set of attributes FK in relation schema R1 is a foreign key of R1 that references relation R2 if it satisfies the following two rules:

1. The attributes in FK have the same domain(s) as the primary key attributes PK of R2; the attributes FK are said to reference or refer to the relation R2.

2. Let $t_1 \in r(R1)$ and $t_2 \in r(R2)$, then

   either t1 [FK] = t2[PK] or t1[FK] = NULL

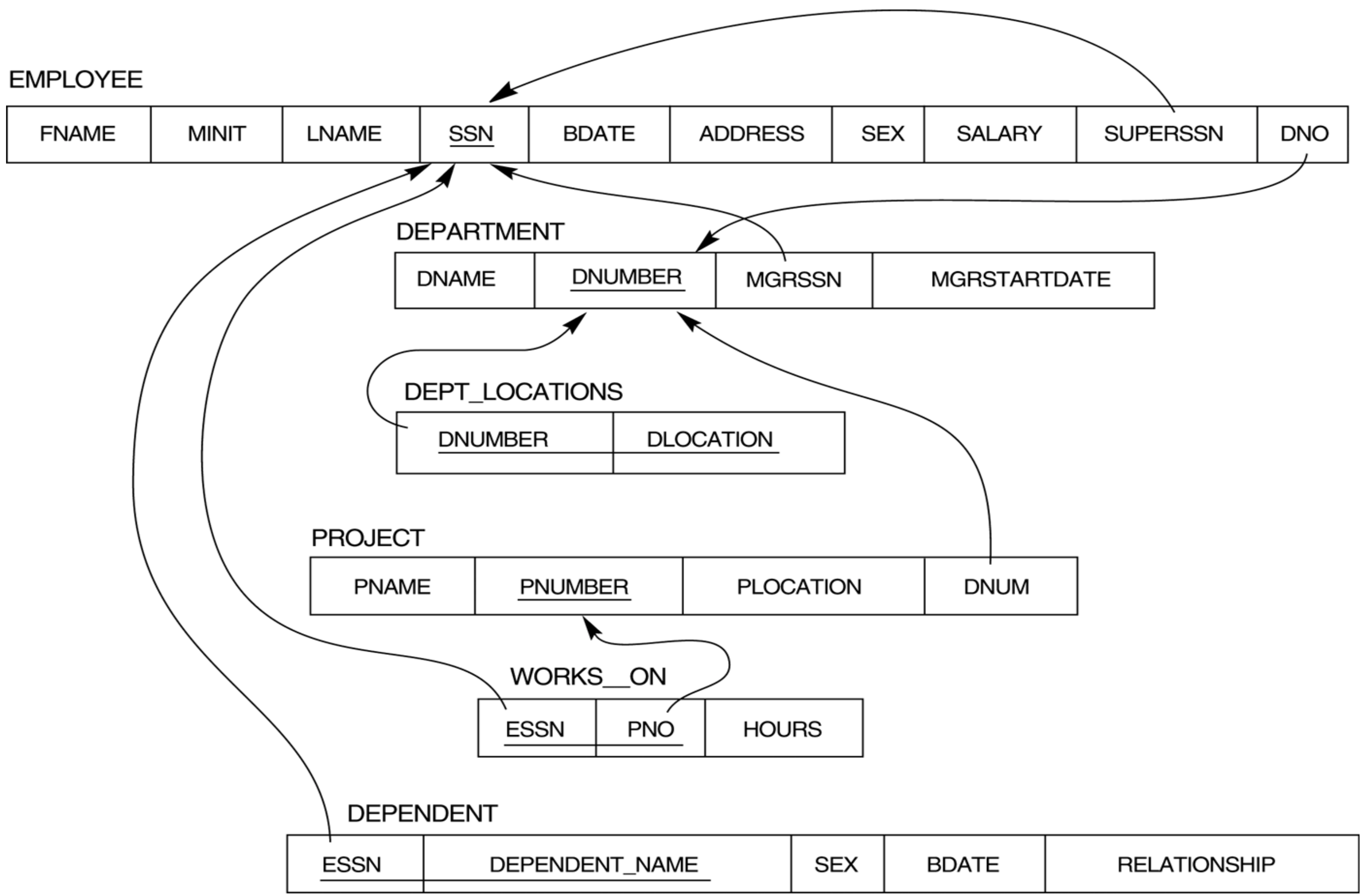
R1 is called the referencing relation and R2 is the referenced relation.

A foreign key can refer to its own relation.

e.g. : SUPERSSN of EMPLOYEE refers to SSN of EMPLOYEE

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

Displaying Referential integrity constraints



Referential integrity constraints displayed on the COMPANY relational database schema.

Other types of constraint

1. Semantic integrity constraints:

- Include a large class of general constraints

  E.g. : Salary of an employ should not exceed the salary of the employee's supervisor.

- Such constraints can be specified and enforced by
  - the application programs or
  - constraint specification language (triggers and assertions)

Other types of constraint

2. Functional Dependency :

- Establishes a functional relationship among two sets of attributes X and Y.
- *It specifies that the value of X determines value of Y in all states of a relation.* ***It is denoted by $X \rightarrow Y$***

E.g: In a **STUDENT** relation, **Regno** determines values of every other attributes in the relation.

3. Transition Constraints : It deals with changes in the database and enforced by application programs

E.g : The salary of an employ can only increase.

3.4 Update Operations and Dealing with Constraint Violations

The operations of the relational model can be categorized into **retrievals** and **updates.**

•**Update Operations:**

1. The Insert Operation
2. The Delete Operation
3. The Update Operation

- **Insert** is used to insert a new tuple or tuples in a relation
- **Delete** is used to delete tuples
- **Update** (Modify) is used to change the values of some attributes in existing tuples

Example Database state

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

3.4 Update Operations and Dealing with Constraint Violations

The Insert Operation

•The **Insert** operation provides a list of attribute values for a new tuple **t** that is to be inserted into a relation **R**.

•Insert can violate any of the four types of constraints

1. Domain constraints.
2. Key constraints
3. Entity integrity
4. Referential integrity

3.4 Update Operations and Dealing with Constraint Violations

The Insert Operation : Examples of violations

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

1. **Insert** <'Cecilia', 'F', 'Kolonsky', **null,** '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, null, 4> into **EMPLOYEE**
2. **Insert** <'Alicia', 'J', 'Zelaya', '**999887777',** '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, '987654321', 4> into **EMPLOYEE**..
3. **Insert** <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', F, 28000, '987654321', **7**> into **EMPLOYEE.**
4. **Insert** <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, null, 4> into **EMPLOYEE.**

3.4 Update Operations and Dealing with Constraint Violations

The Insert Operation :

What DBMS will do when Insert operation violates constraints ?

There are several options for DBMS when insert operations causes violations.

1. *Reject the insertion.*
2. *Attempt to correct reason for rejecting the insertion.*

3.4 Update Operations and Dealing with Constraint Violations

The Delete Operation

The Delete operation can violate only referential integrity, if the tuple being deleted is referenced by the foreign keys from other tuples in the database.

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

3.4 Update Operations and Dealing with Constraint Violations

The Delete Operation :Examples of violations

Are the following delete operations acceptable ?

1. Delete the WORKS_ON tuple with ESSN = ‘999887777’

   and PNO = 10.

2. Delete the EMPLOYEE tuple with SSN = ‘999887777’.

3. Delete the EMPLOYEE tuple with SSN = ‘333445555’.

3.4 Update Operations and Dealing with Constraint Violations

The Delete Operation

What DBMS will do when the Delete Operation violates referential integrity constraints ?

There are several options for DBMS.

1. *Reject the deletion.*
2. *Cascade the deletion.*
3. *Modify the referencing attribute values that causes the violation.*
4. *Combination of all 3 options.*

3.4 Update Operations and Dealing with Constraint Violations

The Update Operation

- The Update operation is used to change the values of one or more attributes in a tuple (or tuples) of some relation R. It is necessary to specify a condition on the attributes of the relation to select the tuple (or tuples) to be modified.

- Updating an attribute that is neither a **primary key** nor a **foreign key** usually causes no problems.

- Modifying a **primary key** value is similar to deleting one tuple and inserting another in its place.

- If a **foreign key** attribute is modified, the DBMS must make sure that the new value refers to an existing tuple in the referenced relation .

3.4 Update Operations and Dealing with Constraint Violations

The Update Operation: Examples of violations

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

3.4 Update Operations and Dealing with Constraint Violations

The Update Operation : Examples of updates and violations

Are the following Update operations acceptable ?

1. Update the SALARY of the EMPLOYEE tuple with SSN = ‘999887777’ to 28000.
2. Update the DNO of the EMPLOYEE tuple with SSN = ‘999887777’ to 1.
3. Update the DNO of the EMPLOYEE tuple with SSN = ‘999887777’ to 7
4. Update the SSN of the EMPLOYEE tuple with SSN = ‘999887777’ to ‘987654321’.

The Relational Algebra

The Relational Algebra

- The basic set of operations for relational model is the **relational algebra**
- The relational algebra operations are usually divided into two groups.
- One group includes set operations from mathematical set theory.
  1. **UNION**
  2. **INTERSECTION**
  3. **SET DIFFERENCE**
  4. **CARTESIAN PRODUCT**

- The other group consists of operations developed specifically for relational databases.
  1. **SELECT (Unary operation)**
  2. **PROJECT (Unary operation)**
  3. **JOIN (Binary operation)**

SELECT

- The SELECT operation is used to select a subset of the tuples from a relation that satisfy a selection condition.
- The SELECT operation can also be visualized as a ***horizontal partition*** of the relation into two sets of tuples
  1. those tuples that satisfy the condition and are selected
  2. those tuples that do not satisfy the condition and are discarded.
- Denoted by σ

SELECT

- **General format:**

$$\sigma_{<\text{selection condition}>}(R)$$

Where

- the symbol σ (sigma) is used to denote the SELECT operator.
- ***selection condition*** is a Boolean expression specified on the attributes of relation R.
- R is generally ***a relational algebra expression*** whose result is a relation.

- *The Relation resulting from the SELECT operation has the same attributes as R*

SELECT

Examples :

1. select the EMPLOYEE tuples whose department is 4,

   $\sigma_{DNO=4}(EMPLOYEE)$

2. select the EMPLOYEE tuples whose salary is greater than 30,000,

   $\sigma_{SALARY>30000}(EMPLOYEE)$

SELECT

- The Boolean expression specified in **<selection condition>** is made up of a number of clauses of the form

1. <attribute name> <comparison operator> <constant value>

   or

2. <attribute name> <comparison operator> < attribute name >

- <attribute name> is the name an attribute of R
- Comparison operator is one of the operators $\{=,<,>,\leq,\geq,\neq\}$

SELECT

- Clauses can be connected arbitrarily using Boolean operators ***AND, OR and NOT***

For example: **S**elect all employees who either work in department 4 and make over 25,000 per year, or work in department 5 and make over 30,000.

$$\sigma_{\text{(DNO=4 AND SALARY>25000) OR (DNO=5 AND SALARY>30000)}}\text{(EMPLOYEE)}$$

SELECT

Employee Relation

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

Results :

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
Franklin	T	Wong	333445555	1955-12-08	638 Voss,Houston,TX	M	40000	888665555	5
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry,Bellaire,TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 FireOak,Humble,TX	M	38000	333445555	5

SELECT

- The Boolean conditions AND, OR, and NOT have their normal interpretation as follows:
  - (cond1 AND cond2) is true if both (cond1) and (cond2) are true; otherwise, it is false.
  - (cond1 OR cond2) is true if either (cond1) or (cond2) or both are true; otherwise, it is false.
  - (NOT cond) is true if cond is false; otherwise, it is false.

SELECT

Note the following

- The number of tuples in the resulting relation is always *less than or equal to* the number of tuples in R.
- The **degree** of the relation resulting from a SELECT operation is the same as that of R.

- The SELECT operation is **commutative;** that is:

$$\sigma_{<cond1>}(\sigma_{<cond2>}(R)) = \sigma_{<cond2>}(\sigma_{<cond1>}(R))$$

- A cascade of select operation can be combined into one SELECT operation using AND condition

$$\sigma_{<cond1>}(\sigma_{<cond2>}(\ldots(\sigma_{<condn>}(R))\ldots)) = \sigma_{<cond1>AND\ <cond2>\ldots AND\ <condn>}(R)$$

PROJECT

- The PROJECT operation selects certain columns from the table and discards the other columns.
- Can be visualized as a vertical partition of the relation into two relations:
  1. one has the needed columns and contains the result of the operation,
  2. and the other contains the discarded columns.

- **Denoted by π**
- **General form :**

$$\pi_{\text{<attribute list>}}(R)$$

PROJECT

E.g. :

list each employee's first and last name and salary,

$$\pi_{\text{LNAME, FNAME, SALARY}}(\textbf{EMPLOYEE})$$

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

Result

LNAME	FNAME	SALARY
Smith	John	30000
Wong	Franklin	40000
Zelaya	Alicia	25000
Wallace	Jennifer	43000
Narayan	Ramesh	38000
English	Joyce	25000
Jabbar	Ahmad	25000
Borg	James	55000

PROJECT

- The PROJECT operation removes any duplicate tuples (a valid relation)

List SEX and SALARY of Employees

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

$$\pi_{\text{SEX, SALARY}}(\text{EMPLOYEE})$$

SEX	SALARY
M	30000
M	40000
F	25000
F	43000
M	38000
M	25000
M	55000

PROJECT

Note the Following :

- The number of tuples in a relation resulting from a PROJECT operation is always less than or equal to the number of tuples in R.
- If the projection list is a superkey of R then resulting relation has the same number of tuples as R

- $\pi_{<list1>}(\pi_{<list2>}(R)) = \pi_{<list1>}(R)$ as long as <list2> contains the attributes in <list1>

E.g : $\pi_{NAME,SEX}(\pi_{NAME,SEX,SALARY}(R)) = \pi_{NAME,SEX}(R)$

Sequence of Relational Operations

- We can write relational algebra expression in ways
  1. A single relational algebra expression by nesting the operations
  2. we divide the above operation into a sequence of simple operations to produce intermediate results.

E.g. : List LNAME, FNAME, SALARY of all employees whose department no =5

$$\pi_{\text{LNAME, FNAME, SALARY}}(\sigma_{\text{DNO = 5}}(\text{EMPLOYEE}))$$

OR

$$\text{DEP5\_EMPS} \leftarrow \sigma_{\text{DNO=5}}(\text{EMPLOYEE})$$

$$\text{RESULT} \leftarrow \pi_{\text{FNAME, LNAME, SALARY}}(\text{DEP5\_EMPS})$$

Result of both the operations

FNAME	LNAME	SALARY
John	Smith	30000
Franklin	Wong	40000
Ramesh	Narayan	38000
Joyce	English	25000

The RENAME operation

: Used rename attributes of a relation.

Example :

$$TEMP \leftarrow \sigma_{DNO = 5}(EMPLOYEE)$$

$$R(FISRTNAME, LASTNAME, SALARY) \leftarrow \pi_{FNAME,\ LNAME,\ SALARY}(TEMP)$$

TEMP	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren,Houston,TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss,Houston,TX	M	40000	888665555	5
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak,Humble,TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000	333445555	5

LNAME, FNAME, SALARY are renamed as FISRTNAME, LASTNAME,SALARY

R	FIRSTNAME	LASTNAME	SALARY
	John	Smith	30000
	Franklin	Wong	40000
	Ramesh	Narayan	38000
	Joyce	English	25000

The RENAME operation

- We can also define a formal RENAME operation-which can rename either the relation name or the attribute names, or both

- The general RENAME operation when applied to a relation R of degree n is denoted by

$$\rho_{S(B_1, B_2, \ldots, B_n)}(R) \text{ or } \rho_S(R) \text{ or } \rho_{(B_1, B_2, \ldots, B_n)}(R)$$

S→ new relation name

Set Theoretic Operations

- **UNION:** The result of this operation, denoted by **R U S**, is a relation that includes all tuples that are either in R or in S or in both R and S. Duplicate tuples are eliminated.

- **INTERSECTION:** The result of this operation, denoted by **R ∩ S,** is a relation that includes all tuples that are in both R and S.

- **SET DIFFERENCE:** The result of this operation, denoted by **R - S,** is a relation that includes all tuples that are in R but not in S.

- **Union (Type) compatibility** :

  Two relations R(A1, A2, . . ., An) and S(B1, B2, . . ., Bn) are said to be union compatible **if they have the same degree n**, and if

  dom(Ai) = dom(Bi) for 1 ≤ i ≤ n.

Set Theoretic Operations

Example

(a)

STUDENT	FN	LN
	Susan	Yao
	Ramesh	Shah
	Johnny	Kohler
	Barbara	Jones
	Amy	Ford
	Jimmy	Wang
	Ernest	Gilbert

INSTRUCTOR	FNAME	LNAME
	John	Smith
	Ricardo	Browne
	Susan	Yao
	Francis	Johnson
	Ramesh	Shah

(b)

FN	LN
Susan	Yao
Ramesh	Shah
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert
John	Smith
Ricardo	Browne
Francis	Johnson

(c)

FN	LN
Susan	Yao
Ramesh	Shah

(d)

FN	LN
Johnny	Kohler
Barbara	Jones
Amy	Ford
Jimmy	Wang
Ernest	Gilbert

(e)

FNAME	LNAME
John	Smith
Ricardo	Browne
Francis	Johnson

Figure 07.11

Illustrating the set operations UNION, INTERSECTION, and DIFFERENCE. (a) Two union compatible relations. (b) STUDENT $\cup$ INSTRUCTOR. (c) STUDENT $\cap$ INSTRUCTOR. (d) STUDENT $-$ INSTRUCTOR. (e) INSTRUCTOR $-$ STUDENT.

Set Theoretic Operations

Note the following

- Both UNION and INTERSECTION are ***commutative operations;*** that is

$$R \cup S = S \cup R, \text{ and } R \cap S = S \cap R$$

- Both union and intersection can be treated as n-ary operations applicable to any number of relations because both are *associative operations;* that is

$$R \cup (S \cup T) = (R \cup S) \cup T, \text{ and } (R \cap S) \cap T = R \cap (S \cap T)$$

- The minus operation is ***not*** *commutative;* that is, in general

$$R - S \neq S - R$$

CARTESIAN (or cross product) Operation

- This operation is used to combine tuples from two relations in a combinatorial fashion.
- In general, the result of $\mathbf{R(A_1, A_2, \ldots, A_n) \times S(B_1, B_2, \ldots, B_m)}$ is a relation **Q** with degree **n + m** attributes $\mathbf{Q(A_1, A_2, \ldots, A_n, B_1, B_2, \ldots, B_m)}$, in that order. The resulting relation Q has one tuple for each combination of tuples—one from R and one from S.
  - Hence, if R has n_R tuples (denoted as $|R| = n_R$), and S has n_S tuples, then

    $\mathbf{| R \times S |}$ will have $\mathbf{n_R * n_S}$ tuples.
  - The two relations do **NOT** have to be **"type compatible"**

CARTESIAN (or cross product) Operation

- **Example for Cross product**

EMPNAMES

FNAME	LNAME	SSN
Alicia	Zelaya	999887777
Jennifer	Wallace	987654321
Joyce	English	453453453

DEPENDENT

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
333445555	Alice	F	1986-04-05	DAUGHTER
333445555	Theodore	M	1983-10-25	SON
333445555	Joy	F	1958-05-03	SPOUSE
987654321	Abner	M	1942-02-28	SPOUSE
123456789	Michael	M	1988-01-04	SON
123456789	Alice	F	1988-12-30	DAUGHTER
123456789	Elizabeth	F	1967-05-05	SPOUSE

CARTESIAN (or cross product) Operation

- **Example for Cross product**

EMP_DEPENDENTS ← EMPNAMES x DEPENDENT

EMP_DEPENDENTS	FNAME	LNAME	SSN	ESSN	DEPENDENT_NAME	SEX	BDATE	• • •
	Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	• • •
	Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	• • •
	Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	• • •
	Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	• • •
	Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	• • •
	Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	• • •
	Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	• • •
	Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	• • •
	Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	• • •
	Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	• • •
	Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	• • •
	Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	• • •
	Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	• • •
	Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	• • •
	Joyce	English	453453453	333445555	Alice	F	1986-04-05	• • •
	Joyce	English	453453453	333445555	Theodore	M	1983-10-25	• • •
	Joyce	English	453453453	333445555	Joy	F	1958-05-03	• • •
	Joyce	English	453453453	987654321	Abner	M	1942-02-28	• • •
	Joyce	English	453453453	123456789	Michael	M	1988-01-04	• • •
	Joyce	English	453453453	123456789	Alice	F	1988-12-30	• • •
	Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	• • •

CARTESIAN (cross product) Operation

- The Cross product is useful when followed by a selection that matches the values of attributes coming from the component relations

For Example : We want to retrieve a list of female employee's dependents

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

CARTESIAN (cross product) Operation

Example : Retrieving a list of female employee's dependents

1. **FEMALE_EMPS $\leftarrow \sigma_{SEX='F'}$(EMPLOYEE)**

FEMALE_EMPS	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle,Spring,TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry,Bellaire,TX	F	43000	888665555	4
	Joyce	A	English	453453453	1972-07-31	5631 Rice,Houston,TX	F	25000	333445555	5

2. **EMPNAMES $\leftarrow \pi_{FNAME,\ LNAME,\ SSN}$ (FEMALE_EMPS)**

EMPNAMES	FNAME	LNAME	SSN
	Alicia	Zelaya	999887777
	Jennifer	Wallace	987654321
	Joyce	English	453453453

CARTESIAN (cross product) Operation

3. EMP_DEPENDENTS ← EMPNAMES x DEPENDENT

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

EMP_DEPENDENTS	FNAME	LNAME	SSN	ESSN	DEPENDENT_NAME	SEX	BDATE	• • •
	Alicia	Zelaya	999887777	333445555	Alice	F	1986-04-05	• • •
	Alicia	Zelaya	999887777	333445555	Theodore	M	1983-10-25	• • •
	Alicia	Zelaya	999887777	333445555	Joy	F	1958-05-03	• • •
	Alicia	Zelaya	999887777	987654321	Abner	M	1942-02-28	• • •
	Alicia	Zelaya	999887777	123456789	Michael	M	1988-01-04	• • •
	Alicia	Zelaya	999887777	123456789	Alice	F	1988-12-30	• • •
	Alicia	Zelaya	999887777	123456789	Elizabeth	F	1967-05-05	• • •
	Jennifer	Wallace	987654321	333445555	Alice	F	1986-04-05	• • •
	Jennifer	Wallace	987654321	333445555	Theodore	M	1983-10-25	• • •
	Jennifer	Wallace	987654321	333445555	Joy	F	1958-05-03	• • •
	Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	• • •
	Jennifer	Wallace	987654321	123456789	Michael	M	1988-01-04	• • •
	Jennifer	Wallace	987654321	123456789	Alice	F	1988-12-30	• • •
	Jennifer	Wallace	987654321	123456789	Elizabeth	F	1967-05-05	• • •
	Joyce	English	453453453	333445555	Alice	F	1986-04-05	• • •
	Joyce	English	453453453	333445555	Theodore	M	1983-10-25	• • •
	Joyce	English	453453453	333445555	Joy	F	1958-05-03	• • •
	Joyce	English	453453453	987654321	Abner	M	1942-02-28	• • •
	Joyce	English	453453453	123456789	Michael	M	1988-01-04	• • •
	Joyce	English	453453453	123456789	Alice	F	1988-12-30	• • •
	Joyce	English	453453453	123456789	Elizabeth	F	1967-05-05	• • •

CARTESIAN (cross product) Operation

4. ACTUAL_DEPENDENTS $\leftarrow \sigma_{SSN=ESSN}$ (EMP_DEPENDENTS)

ACTUAL_DEPENDENTS	FNAME	LNAME	SSN	ESSN	DEPENDENT_NAME	SEX	BDATE	• • •
	Jennifer	Wallace	987654321	987654321	Abner	M	1942-02-28	• • •

5. RESULT $\leftarrow \pi_{\text{FNAME, LNAME, DEPEDENT\_NAME}}$(ACTUAL_DEPENDENTS)

RESULT	FNAME	LNAME	DEPENDENT_NAME
	Jennifer	Wallace	Abner

JOIN Operation

- The sequence of Cartesian product followed by a SELECT is used quite commonly to identify and select related tuples from two relations, a special operation, called **JOIN**.

- This operation is very important for any relational database with more than a single relation, because it allows us to process relationships among relations.

- Join is denoted by $\bowtie$

JOIN Operation

The General Form

- A join operation on two relations **R(A_1, A_2, . . ., A_n)** and **S(B_1, B_2, . . ., B_m)** is:

$$R \bowtie_{\text{<join condition>}} S$$

- **R** and **S** can be any relations that result from general *relational algebra expressions.*
- The relation will have (n + m) attributes
- A1,A2,…An,B1,B2…Bm are the attributes of the resultant relation in that order

JOIN Operation

Example: Retrieve the name of the manager of each department.

To get the manager's name, we need to combine each DEPARTMENT tuple with the EMPLOYEE tuple whose SSN value matches the MGRSSN value in the department tuple. We do this by using the join $\bowtie$ operation.

$$\textbf{DEPT\_MGR} \leftarrow \textbf{DEPARTMENT} \bowtie_{\textbf{MGRSSN=SSN}} \textbf{EMPLOYEE}$$

Note : *The above operation is equivalent to Cartesian product of DEPARTMENT and EMPLOYEE followed by SELECT operation on the resultant relation , with the condition MGRSSN=SSN*

$$\textbf{EMP\_MANG} \leftarrow \textbf{EMPLOYEE} \times \textbf{DEPARTMENT}$$

$$\textbf{DEPT\_MGR} \leftarrow \sigma_{\textbf{SSN=MGRSSN}} (\textbf{EMP\_MANG})$$

JOIN Operation

JOIN Operation : DEPT_MGR $\leftarrow$ DEPARTMENT $\bowtie_{\text{MGRSSN=SSN}}$ EMPLOYEE

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

DEPT_MGR	DNAME	DNUMBER	MGRSSN	• • •	FNAME	MINIT	LNAME	SSN	• • •
	Research	5	333445555	• • •	Franklin	T	Wong	333445555	• • •
	Administration	4	987654321	• • •	Jennifer	S	Wallace	987654321	• • •
	Headquarters	1	888665555	• • •	James	E	Borg	888665555	• • •

Figure 07.13
Illustrating the JOIN operation.

JOIN Operation

Note the following

- In JOIN, only combinations of tuples *satisfying the join condition* appear in the result .
- Whereas in the CARTESIAN PRODUCT *all combinations of tuples* are included in the result

- **THETA join :**

  <condition> AND <condition> AND . . . AND <condition>

  Where each condition is of the form $A_i \ \Theta \ B_j$,

  A_i is an attribute of R, B_j is an attribute of S (same domain) and

  $\Theta = \{ =,<,>,\leq,\geq,\neq \}$.

The EQUIJON and NATURAL JOIN:

- A JOIN, where the only comparison operator used is **=**, is called an **EQUIJOIN.**
- In the result of an EQUIJOIN we always have one or more pairs of attributes that have *identical values* in every tuple. To avoid the second attribute we use **NATURAL JOIN.**
- NATURAL JOIN requires that the two join attributes (or each pair of join attributes) ***have the same name in both relations.***
- *To have same attribute name in both the relations , must rename one of the two attributes using rename operator (ρ).*

- **NATURAL JOIN** of **R** and **S** is denoted by

  R * S

3.5 The Relational Algebra

<u>Binary Relational Operations</u> : JOIN and DIVISION

The EQUIJON and NATURAL JOIN: Example

PROJ_DEPT = PROJECT * $\rho_{\text{(DNAME, DNUM,MGRSSN,MGRSTARTDATE)}}$(DEPARTMENT)

DNUMBER of DEPARTMENT is renamed as DNUM

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

In the resultant relation only one DNUM is selected and displayed.

PROJ_DEPT	PNAME	PNUMBER	PLOCATION	DNUM	DNAME	MGRSSN	MGRSTARTDATE
	ProductX	1	Bellaire	5	Research	333445555	1988-05-22
	ProductY	2	Sugarland	5	Research	333445555	1988-05-22
	ProductZ	3	Houston	5	Research	333445555	1988-05-22
	Computerization	10	Stafford	4	Administration	987654321	1995-01-01
	Reorganization	20	Houston	1	Headquarters	888665555	1981-06-19
	Newbenefits	30	Stafford	4	Administration	987654321	1995-01-01

More about JOIN

- A more general but non-standard definition for NATURAL JOIN is

$$Q = R *_{(<list1>),(<list2>)} S$$

- In general, if R has n_R tuples and S has n_S tuples, the result of a JOIN operation $R \bowtie_{<join\ condition>} S$ will have between zero and $n_R * n_S$ tuples.

- The expected size of the join result divided by the maximum size $n_R * n_S$ leads to a ratio called **join selectivity**, which is a property of each join condition.

More about JOIN

- If there is no join condition, all combinations of tuples qualify and the JOIN becomes a **CARTESIAN PRODUCT**, also called **CROSS PRODUCT or CROSS JOIN**.

- The natural join or **EQUIJON** operation can also be specified among multiple tables, leading to an n-way join.

$$((\text{PROJECT} \bowtie_{\text{DNUM=DNUMBER}} \text{DEPARTMENT}) \bowtie_{\text{MGRSSN=SSN}} \text{EMPLOYEE})$$

Complete Set of Relational Operations

- The set of operations { **σ, π , ∪, - , X}** is called a **complete set** because any other relational algebra expression can be expressed by a combination of these five operations.

For example:

$$\mathbf{R \cap S \equiv (R \cup S) - ((R - S) \cup (S - R))}$$

$$R \bowtie_{\text{<join condition>}} S \equiv \sigma_{\text{<condition>}} (\mathbf{R\ X\ S})$$

DIVISION

Let Z be the attributes of R

Let X be the attributes of S and ***X is subset of Z.***

Then division operation of R and S is denoted by :**T(Y) = R(Z) ÷ S(X),**

where the attributes in the resultant relation T are the attributes of R that are not attributes of S.

i.e Y = Z - X (and hence Z = X ∪ Y)

AND ,

the tuples of the resultant relation **T** are such that ***each tuple of T must appear in R in combination with every tuple in S.***

DIVISION

DIVISION : Example

T = R ÷ S

Attributes of T = attr. Of R - attr. Of S

= {A,B} - {A}

= { B}

(b)

R	A	B
	a1	b1
	a2	b1
	a3	b1
	a4	b1
	a1	b2
	a3	b2
	a2	b3
	a3	b3
	a4	b3
	a1	b4
	a2	b4
	a3	b4

S	A
	a1
	a2
	a3

T	B
	b1
	b4

In general , the result of DIVISION is a relation T(Y) that includes a tuple t if tuples t_R appear in R with t_R [Y] = t, and with t_R [X] = t_s *for every tuple* t_s in S.

DIVISION

Retrieve the names of employees who work on ***all*** the projects

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

$$T1(SSN,PNO) \leftarrow \pi_{ESSN,PNO}\ (WORKS\_ON)$$

$$T2 \leftarrow \pi_{PNO}\ (WORKS\_ON)$$

$$T3 \leftarrow (T1 \div T2)$$

$$RESULT \leftarrow \pi_{FNAME,MINT,LNAME}\ (T3 * EMPLOYEE)$$

DIVISION

Retrieve the names of employees who work on ***all*** the projects that john smith works on .

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

DIVISION

Solution :

Step 1 : Retrieve the list project numbers that john smith works on and store those project numbers in SMITH_PNOS temporary relation.

SMITH $\leftarrow$ $\sigma_{\textbf{FNAME = 'John' AND LNAME = 'Smith'}}$ **(EMPLOYEE)**

SMITH_PNOS $\leftarrow$ $\pi_{\textbf{PNO}}$ (**WORKS_ON** $\bowtie_{\textbf{ESSN=SSN}}$ **SMITH**)

SMITH_PNOS	PNO
	1
	2

DIVISION

:

Step 2 : Select **ESSN** and **PNO**   tuples from **WORKS_ON** relation and store in SSN_PNOS relation.

$$SSN\_PNOS \leftarrow \pi_{\mathbf{ESSN},\ \mathbf{PNO}} (\ \mathbf{WORKS\_ON})$$

SSN_PNOS	ESSN	PNO
	123456789	1
	123456789	2
	666884444	3
	453453453	1
	453453453	2
	333445555	2
	333445555	3
	333445555	10
	333445555	20
	999887777	30
	999887777	10
	987987987	10
	987987987	30
	987654321	30
	987654321	20
	888665555	20

DIVISION

Step 3 : Finally, apply DIVISION operation to the two relation, which gives the desired employees (SSN) working on all the projects that john smith works on.

SMITH_PNOS	PNO
	1
	2

SSN_PNOS	ESSN	PNO
	123456789	1
	123456789	2
	666884444	3
	453453453	1
	453453453	2
	333445555	2
	333445555	3
	333445555	10
	333445555	20
	999887777	30
	999887777	10
	987987987	10
	987987987	30
	987654321	30
	987654321	20
	888665555	20

SSNS(SSN) ← SSN_PNOS ÷ SMITH_PNOS

SSNS	SSN
	123456789
	453453453

DIVISION

We can then select Names of the employees with these SSN by using natural JOIN of SSNS and EMPLOYESS and selecting the desired columns.

$$RESULT \leftarrow \pi_{FNAME\,,\,LNAME}\,(SSNS * EMPLOYEE)$$

So following is the sequence of relation expressions for the query

1. **SMITH** $\leftarrow \sigma_{FNAME = \text{'John'} \text{ AND } LNAME = \text{'Smith'}}$ **(EMPLOYEE)**
2. **SMITH_PNOS** $\leftarrow \pi_{PNO}$ (**WORKS_ON** $\bowtie_{ESSN=SSN}$ **SMITH**)
3. **SSN_PNOS** $\leftarrow \pi_{ESSN\,,\,PNO}$ (**WORKS_ON)**
4. **SSNS(SSN)** $\leftarrow$ **SSN_PNOS** $\div$ **SMITH_PNOS**
5. **RESULT** $\leftarrow \pi_{FNAME\,,\,LNAME}$ **(SSNS * EMPLOYEE)**

Aggregate Functions

- Include Common functions applied to collections of numeric values.

  SUM, AVERAGE, MAXIMUM, and MINIMUM. and COUNT

- An AGGREGATE FUNCTION operation is defined using the symbol

  $\mathcal{F}$ (pronounced "**script F**")

Example: Find the no. of employees and avg salary of employees

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

$\mathcal{F}$ **COUNT** $_{\textbf{SSN}}$**,** **AVERAGE** $_{\textbf{SALARY}}$**(EMPLOYEE)**

COUNT_SSN	AVERAGE_SALARY
8	35125

Aggregate Functions

$\mathcal{F}_{\textbf{MAX}\ \textit{Salary}}$ **(Employee**) retrieves the maximum salary value from the Employee relation (55000)

$\mathcal{F}_{\textbf{MIN}\ \textit{Salary}}$ **(Employee**) retrieves the minimum Salary value from the Employee relation (25000)

$\mathcal{F}_{\textbf{SUM}\ \textit{Salary}}$ **(Employee**) retrieves the sum of the Salary from the Employee relation

Grouping

- Groups the tuples in a relation by the value of some of their attributes and then applying an aggregate function independently to each group.

General form :

$$\boxed{{}_{\text{<grouping attributes>}}\,\mathcal{F}_{\text{<function list>}}\,(R)}$$

Grouping

- **Example :** To group employees by DNO (department number) and compute the count of employees and average salary per department.

$$_{DNO}\mathcal{F}_{COUNT\ SSN,\ AVERAGE\ Salary}(\text{EMPLOYEE})$$

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DNO	COUNT_SSN	AVERAGE_SALARY
5	4	33250
4	3	31000
1	1	55000

Note : duplicates are *not eliminated* when an aggregate function is applied

Examples of Queries on Company database

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q1: Retrieve the name and address of all employees who work for the 'Research' department.

$$\text{RESEARCH\_DEPT} \leftarrow \sigma_{\text{DNAME='Research'}}(\text{DEPARTMENT})$$

$$\text{RESEARCH\_EMPS} \leftarrow (\text{RESEARCH\_DEPT} \bowtie_{\text{DNUMBER= DNO}} \text{EMPLOYEE})$$

$$\text{RESULT} \leftarrow \pi_{\text{FNAME, LNAME, ADDRESS}}(\text{RESEARCH\_EMPS})$$

Examples of Queries on Company database

EMPLOYEE

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS

DNUMBER	DLOCATION
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

DEPARTMENT

DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

WORKS_ON

ESSN	PNO	HOURS
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	null

PROJECT

PNAME	PNUMBER	PLOCATION	DNUM
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
333445555	Alice	F	1986-04-05	DAUGHTER
333445555	Theodore	M	1983-10-25	SON
333445555	Joy	F	1958-05-03	SPOUSE
987654321	Abner	M	1942-02-28	SPOUSE
123456789	Michael	M	1988-01-04	SON
123456789	Alice	F	1988-12-30	DAUGHTER
123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q2: For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birthrate

$$\text{STAFFORD\_PROJS} = \sigma_{\text{PLOCATION='Stafford'}}(\text{PROJECT})$$

$$\text{CONTR\_DEPT} = (\text{STAFFORD\_PROJS} \bowtie_{\text{DNUM=DNUMBER}} \text{DEPARTMENT})$$

$$\text{PROJ\_DEPT\_MGR} = (\text{CONTR\_DEPT} \bowtie_{\text{MGRSSN=SSN}} \text{EMPLOYEE})$$

$$\text{RESULT} = \pi_{\text{PNUMBER, DNUM, LNAME, ADDRESS, BDATE}}(\text{PROJ\_DEPT\_MGR})$$

Examples of Queries on Company database

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q3: Find the names of employees who work on all the projects controlled by department number 5.

$$\text{DEPT5\_PROJS(PNO)} = \pi_{\text{PNUMBER}}(\sigma_{\text{DNUM}=5}(\text{PROJECT}))$$

$$\text{EMP\_PRJO(SSN, PNO)} = \pi_{\text{ESSN, PNO}}(\text{WORKS\_ON})$$

$$\text{RESULT\_EMP\_SSNS} = \text{EMP\_PRJO} \div \text{DEPT5\_PROJS}$$

$$\text{RESULT} = \pi_{\text{LNAME, FNAME}}(\text{RESULT\_EMP\_SSNS} * \text{EMPLOYEE})$$

Examples of Queries on Company database

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q4: Make a list of project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project.

$$SMITHS(ESSN) = \pi_{SSN}(\sigma_{LNAME='Smith'}(EMPLOYEE))$$

$$SMITH\_WORKER\_PROJ = \pi_{PNO}(WORKS\_ON * SMITHS)$$

$$MGRS = \pi_{LNAME,\ DNUMBER}(EMPLOYEE \bowtie_{SSN=MGRSSN} DEPARTMENT)$$

$$SMITH\_MANAGED\_DEPTS(DNUM) = \pi_{DNUMBER}(\sigma_{LNAME='Smith'}(MGRS))$$

$$SMITH\_MGR\_PROJS(PNO) = \pi_{PNUMBER}(SMITH\_MANAGED\_DEPTS * PROJECT)$$

$$RESULT = (SMITH\_WORKER\_PROJS \cup SMITH\_MGR\_PROJS)$$

Examples of Queries on Company database

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q5: List the names of all employees with two or more dependents.

$$T1_{(SSN,\ NO\_OF\_DEPTS)} = {}_{ESSN}\,\mathcal{F}\ {}_{COUNT\ DEPENDENT\_NAME}(DEPENDENT)$$

$$T2 = \sigma_{NO\_OF\_DEPS \geq 2}\ (T1)$$

$$RESULT = \pi_{LNAME,\ FNAME}\ (T2 * EMPLOYEE)$$

Examples of Queries on Company database

EMPLOYEE

FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS

DNUMBER	DLOCATION
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

DEPARTMENT

DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

WORKS_ON

ESSN	PNO	HOURS
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	null

PROJECT

PNAME	PNUMBER	PLOCATION	DNUM
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
333445555	Alice	F	1986-04-05	DAUGHTER
333445555	Theodore	M	1983-10-25	SON
333445555	Joy	F	1958-05-03	SPOUSE
987654321	Abner	M	1942-02-28	SPOUSE
123456789	Michael	M	1988-01-04	SON
123456789	Alice	F	1988-12-30	DAUGHTER
123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q6: Retrieve the names of employees who have no dependents.

$$ALL\_EMPS \leftarrow \pi_{SSN}(EMPLOYEE)$$

$$EMPS\_WITH\_DEPS(SSN) \leftarrow \pi_{ESSN}(DEPENDENT)$$

$$EMPS\_WITHOUT\_DEPS \leftarrow (ALL\_EMPS - EMPS\_WITH\_DEPS)$$

$$RESULT \leftarrow \pi_{LNAME,\ FNAME}(EMPS\_WITHOUT\_DEPS * EMPLOYEE)$$

Examples of Queries on Company database

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

Examples of Queries on Company database

Q7: List the names of managers who have at least one dependent.

$$MGRS(SSN) = \pi_{MGRSSN}(DEPARTMENT)$$

$$EMPS\_WITH\_DEPS(SSN) = \pi_{ESSN}(DEPENDENT)$$

$$MGRS\_WITH\_DEPS = (MGRS \cap EMPS\_WITH\_DEPS)$$

$$RESULT = \pi_{LNAME,\ FNAME}(MGRS\_WITH\_DEPS * EMPLOYEE)$$

Recursive Closure Operations

- This operation is applied to a recursive relationship between tuples of the same type, such as the relationship between an employee and a supervisor.

Example : **To retrieve the SSNs of all employees directly supervised—*at level one*—by the employee e whose name is 'James Borg'**

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

Recursive Closure Operations

To retrieve the SSNs of all employees directly supervised—*at level one*—by the employee e whose name is 'James Borg'

$BORG\_SSN = \pi_{SSN}(\sigma_{FNAME='James' \text{ AND } LNAME='Borg'}(EMPLOYEE))$

$SUPERVISION(SSN1, SSN2) = \pi_{SSN,\ SUPERSSN}(EMPLOYEE)$

$RESULT1(SSN) = \pi_{SSN1}(SUPERVISION \bowtie_{SSN2=SSN} BORG\_SSN)$ (At **level 1**)

(Borg's SSN is 888665555)

SUPERVISION	SSN1 (SSN)	SSN2 (SUPERSSN)
	123456789	333445555
	333445555	888665555
	999887777	987654321
	987654321	888665555
	666884444	333445555
	453453453	333445555
	987987987	987654321

RESULT 1	SSN
	333445555
	987654321

(Supervised by Borg)

Recursive Closure Operations

To retrieve the SSNs of all employees supervised—*at level two* —by 'James Borg' ,

repeat the same operation

RESULT2(SSN) = π_{SSN1} (SUPERVISION $\bowtie_{SSN2=SSN}$ RESULT1) **(level 2)**

(Borg's SSN is 888665555)

SUPERVISION	SSN1 (SSN)	SSN2 (SUPERSSN)
	123456789	333445555
	333445555	888665555
	999887777	987654321
	987654321	888665555
	666884444	333445555
	453453453	333445555
	987987987	987654321

RESULT 1	SSN
	333445555
	987654321

(Supervised by Borg)

RESULT 2	SSN
	123456789
	999887777
	666884444
	453453453
	987987987

(Supervised by Borg's subordinates)

Recursive Closure Operations

To get both sets of employees supervised at levels 1 and 2 by 'James Borg,'

$$\text{RESULT} \leftarrow \text{RESULT2} \cup \text{RESULT1}$$

RESULT	SSN
	123456789
	999887777
	666884444
	453453453
	987987987
	333445555
	987654321

(RESULT1 $\cup$ RESULT2)

3.7 Relational Data base design using ER-to-Relational Mapping Algorithm

ER-to-Relational Mapping Algorithm

Step 1: Mapping of Regular Entity Types

Step 2: Mapping of Weak Entity Types

Step 3: Mapping of Binary 1:1 Relation Types

Step 4: Mapping of Binary 1:N Relationship Types.

Step 5: Mapping of Binary M:N Relationship Types.

Step 6: Mapping of Multivalued attributes.

Step 7: Mapping of N-ary Relationship Types.

FIGURE 7.1
The ER conceptual schema diagram for the COMPANY database.

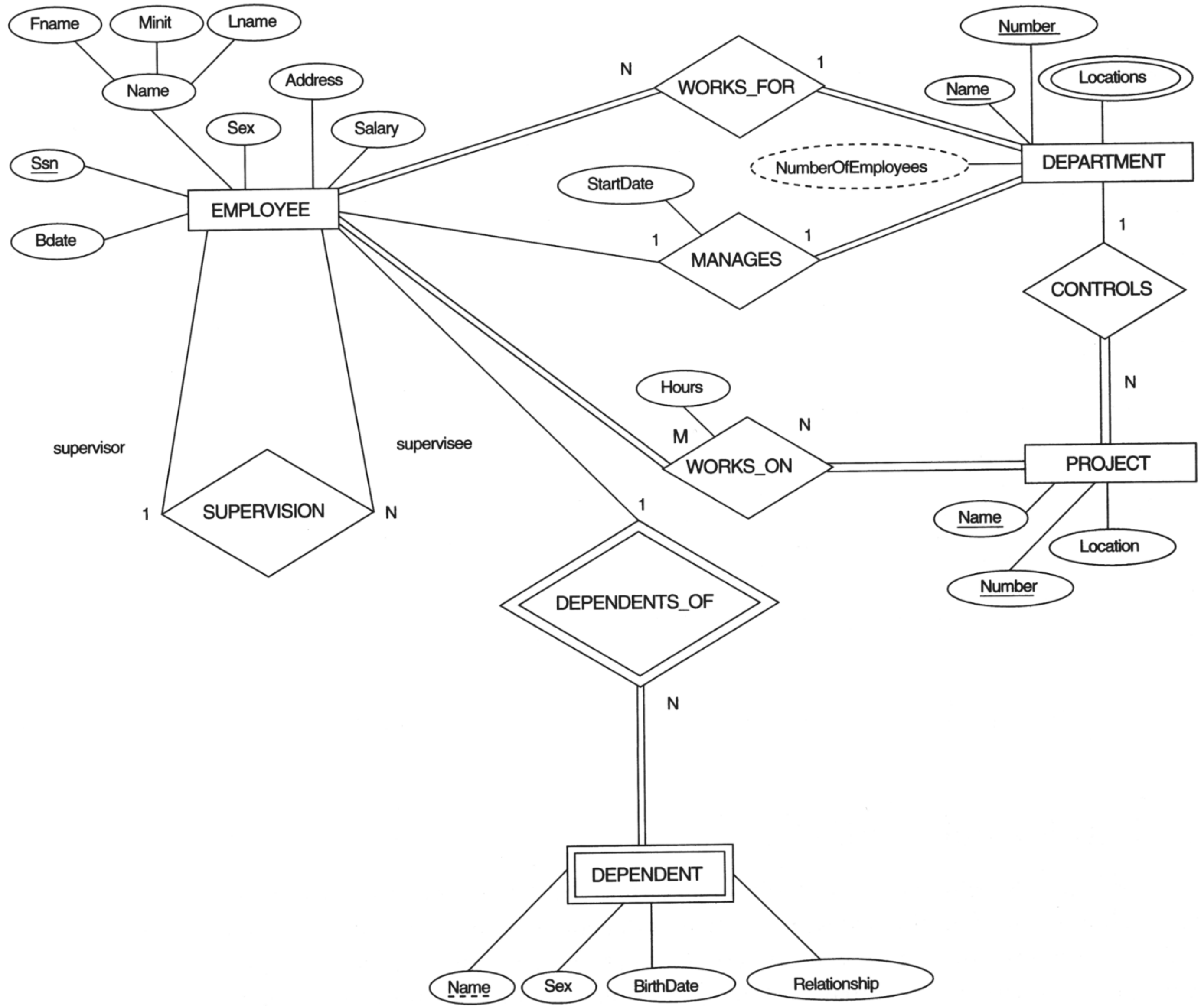
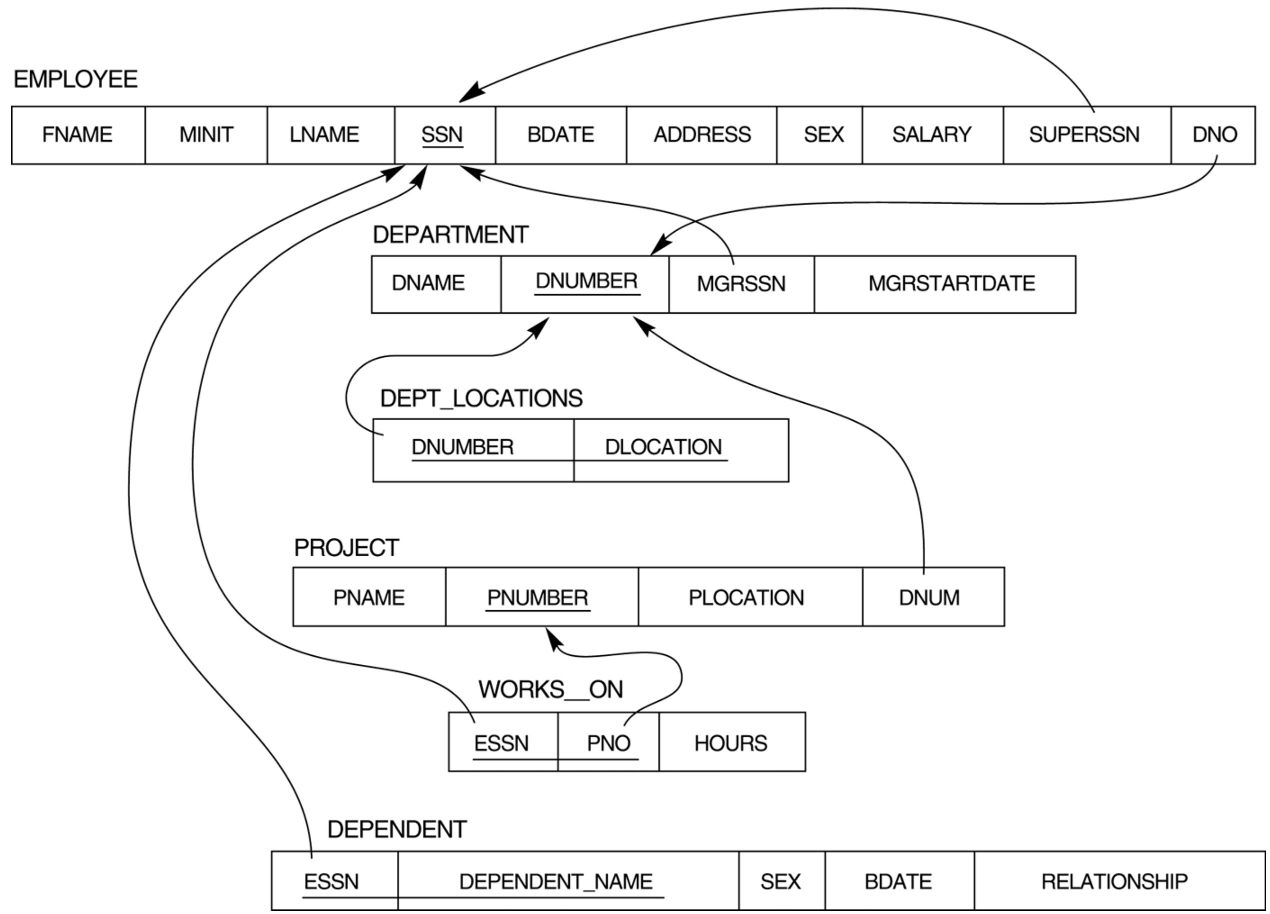


FIGURE 7.2
Result of mapping the COMPANY ER schema into a relational schema.



Step 1: Mapping of Regular Entity Types.

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
- Choose one of the key attributes of E as the primary key for R. If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.

EMPLOYEE

FNAME	MNAME	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY

DEPARTMENT

DNAME	<u>DNUMBER</u>

PROJECT

PNAME	<u>PNUMBER</u>	PLOCATION

Step 2: Mapping of Weak Entity Types

- For each weak entity type W in the ER schema with owner entity type E, create a relation R and include all simple attributes (or simple components of composite attributes) of W as attributes of R.
- In addition, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- The primary key of R is the *combination of* the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

EMPLOYEE

FNAME	MNAME	LNAME	<u>SSN</u>	BDATE	ADDRESS	SEX	SALARY

DEPARTMENT

DNAME	<u>DNUMBER</u>

PROJECT

PNAME	<u>PNUMBER</u>	PLOCATION

DEPNDENT

<u>ESSN</u>	<u>DEPENDENT_NAME</u>	SEX	BDATE	RELATIONSHIP

Step 3: Mapping of Binary 1:1 Relation Types

For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R.

- **There are three possible approaches:**

1. Foreign Key approach

2. Merged relation option

3. Cross-reference or relationship relation option:

Step 3: Mapping of Binary 1:1 Relation Types

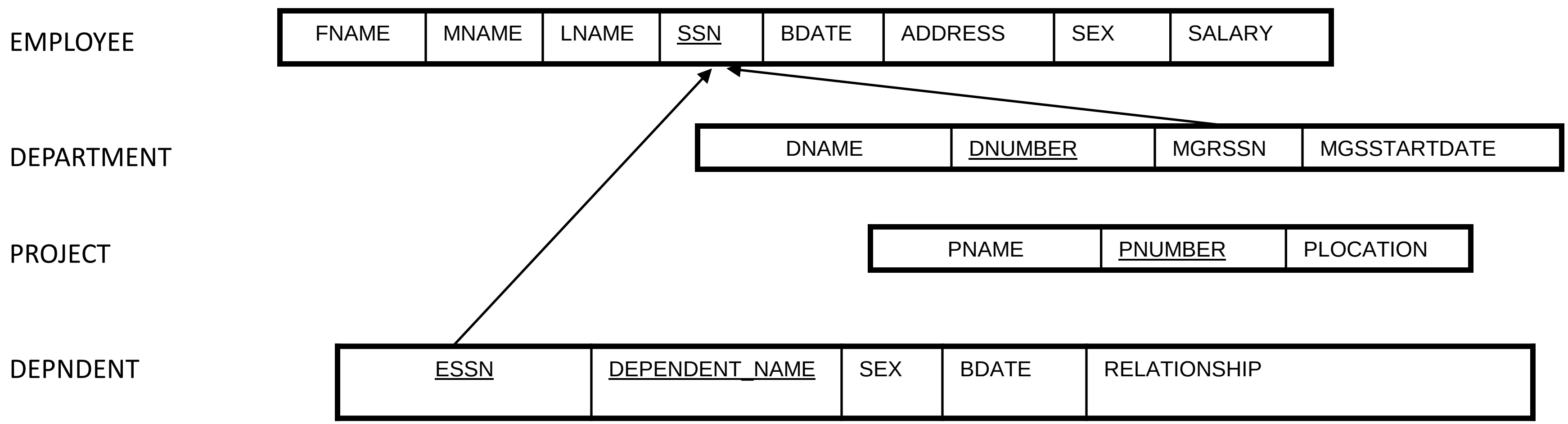
Foreign Key approach:

- Choose one of the relations-S, say-and include as a foreign key in S the primary key of T. It is better to choose an entity type with *total participation* in R in the role of S. Include all the simple attributes of the 1:1 relationship type R as attributes of S.

Example: 1:1 relation MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.

Step 3: Mapping of Binary 1:1 Relation Types

Step 3 RESULT

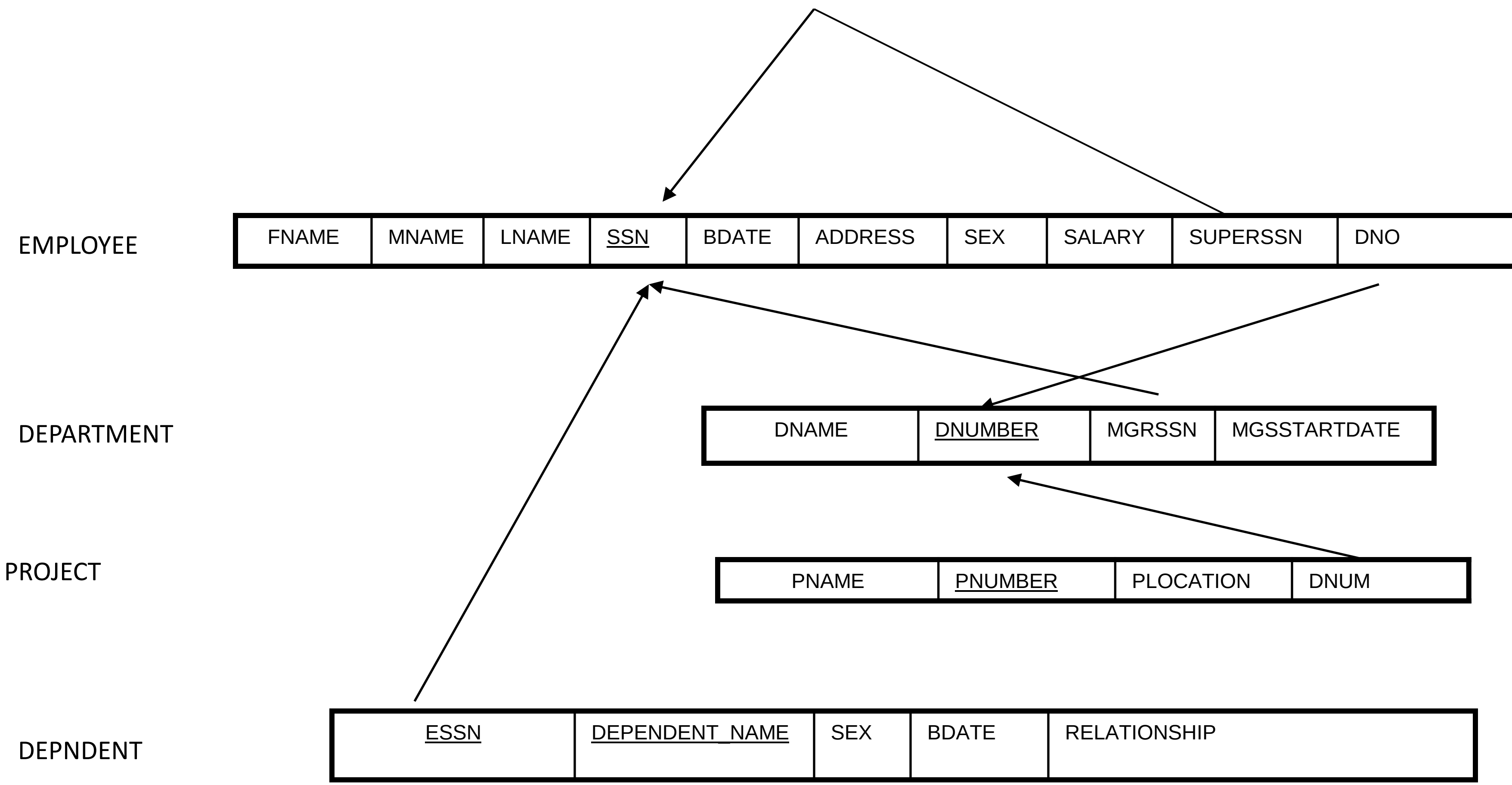


Step 4: Mapping of Binary 1:N Relationship Types.

- For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
- Include any simple attributes of the 1:N relation type as attributes of S.

Step 4: Mapping of Binary 1:N Relationship Types.

- **Step 4:**

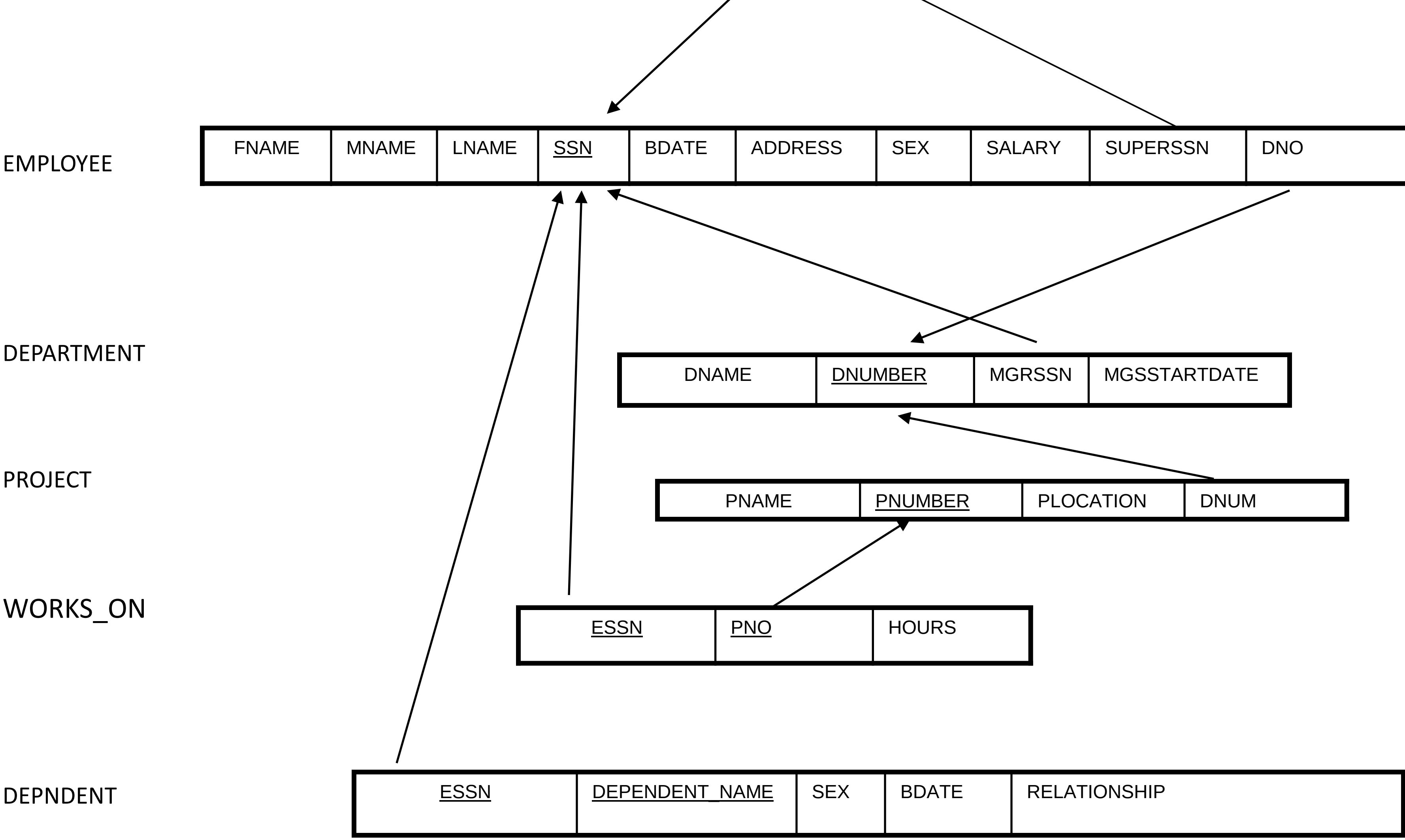


Step 5: Mapping of Binary M:N Relationship Types.

- For each regular binary M:N relationship type R, *create a new relation* S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.

Step 5: Mapping of Binary M:N Relationship Types

STEP : 5 Result



Step 6: Mapping of Multivalued attributes

- For each multivalued attribute A, create a new relation R. This relation R will include an attribute corresponding to A, plus the primary key attribute K-as a foreign key in R-of the relation that represents the entity type of relationship type that has A as an attribute.

– The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

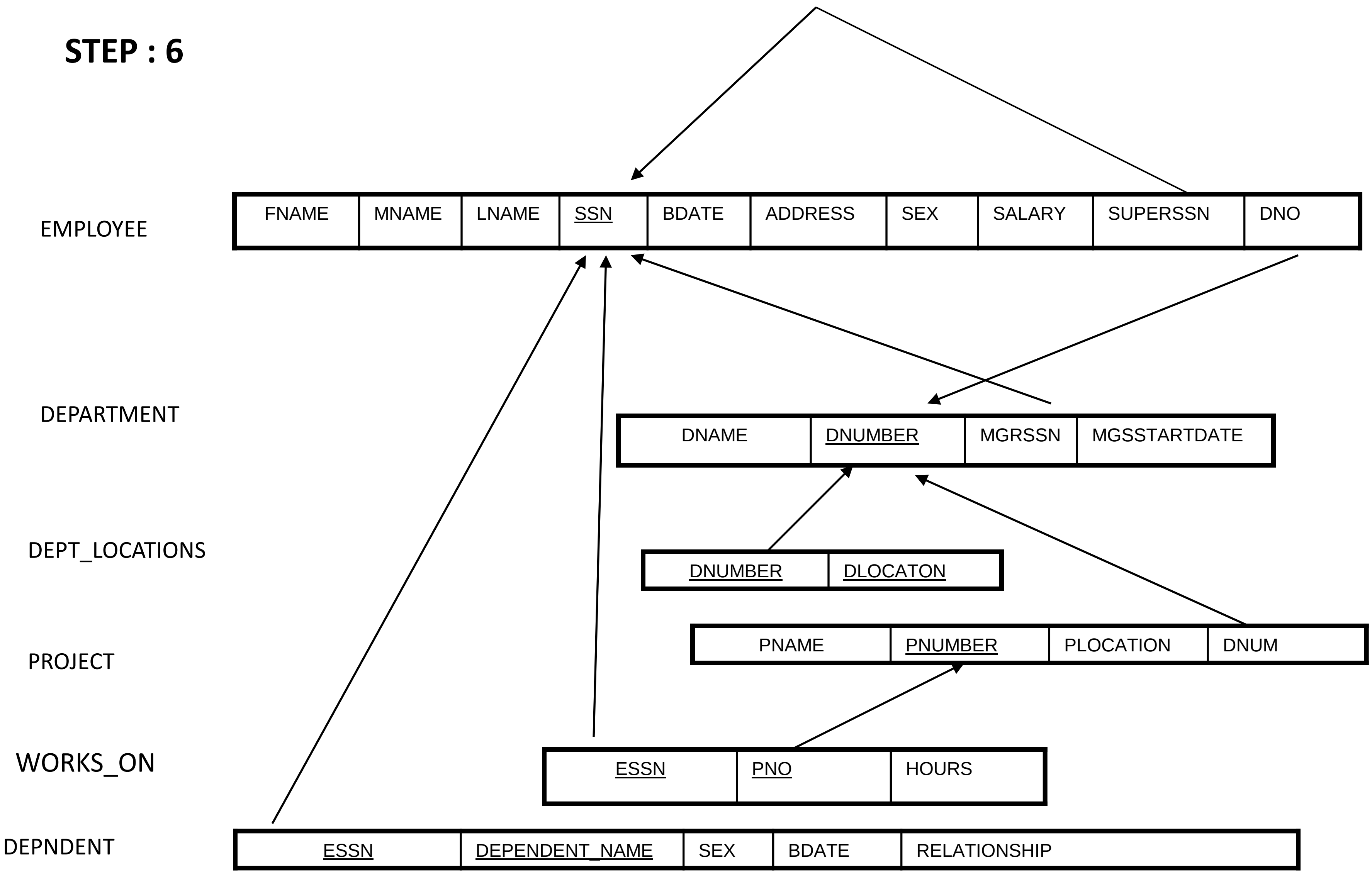
Step 6: Mapping of Multivalued attributes

-

Example: The relation DEPT_LOCATIONS is created. The attribute DLOCATION represents the multivalued attribute LOCATIONS of DEPARTMENT, while DNUMBER-as foreign key-represents the primary key of the DEPARTMENT relation. The primary key of R is the combination of {DNUMBER, DLOCATION}.

Step 6: Mapping of Multivalued attributes

STEP : 6



Step 7: Mapping of N-ary Relationship Types

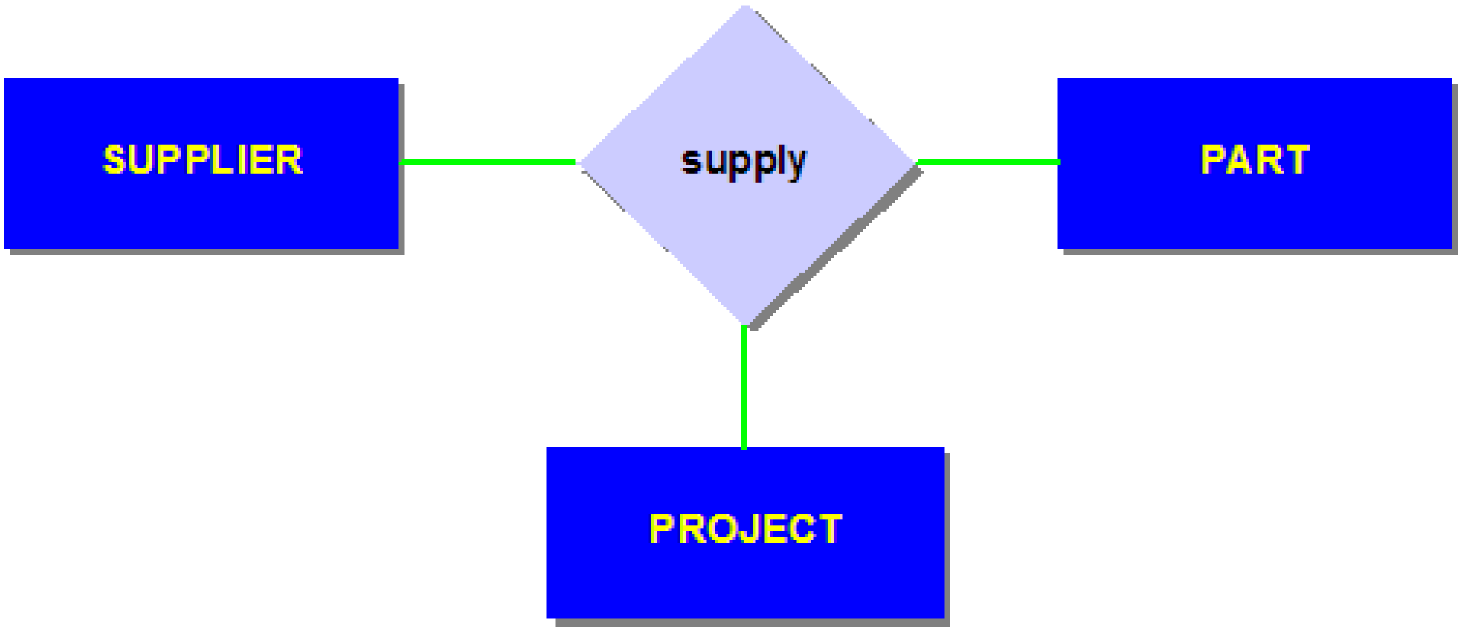
– For each n-ary relationship type R, where $n>2$, create a new relationship S to represent R.

– Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.

– Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.

Step 7: Mapping of N-ary Relationship Types.

Example



Assignment – II

Specify the following queries on the company database schema, using the relational operators.

a. Retrieve the names of all employees in department 5 who work more than 10 hours per week on the ‘ProductX’ project.

b. List the names of all employees who have a dependent with the same first name as themselves.

c. Find the names of all employees who are directly supervised by ‘Franklin Wong’.

d. For each project, list the project name and the total hours per week (by all employees) spent on that project.

e. Retrieve the names of all employees who work on every project.

Assignment - II

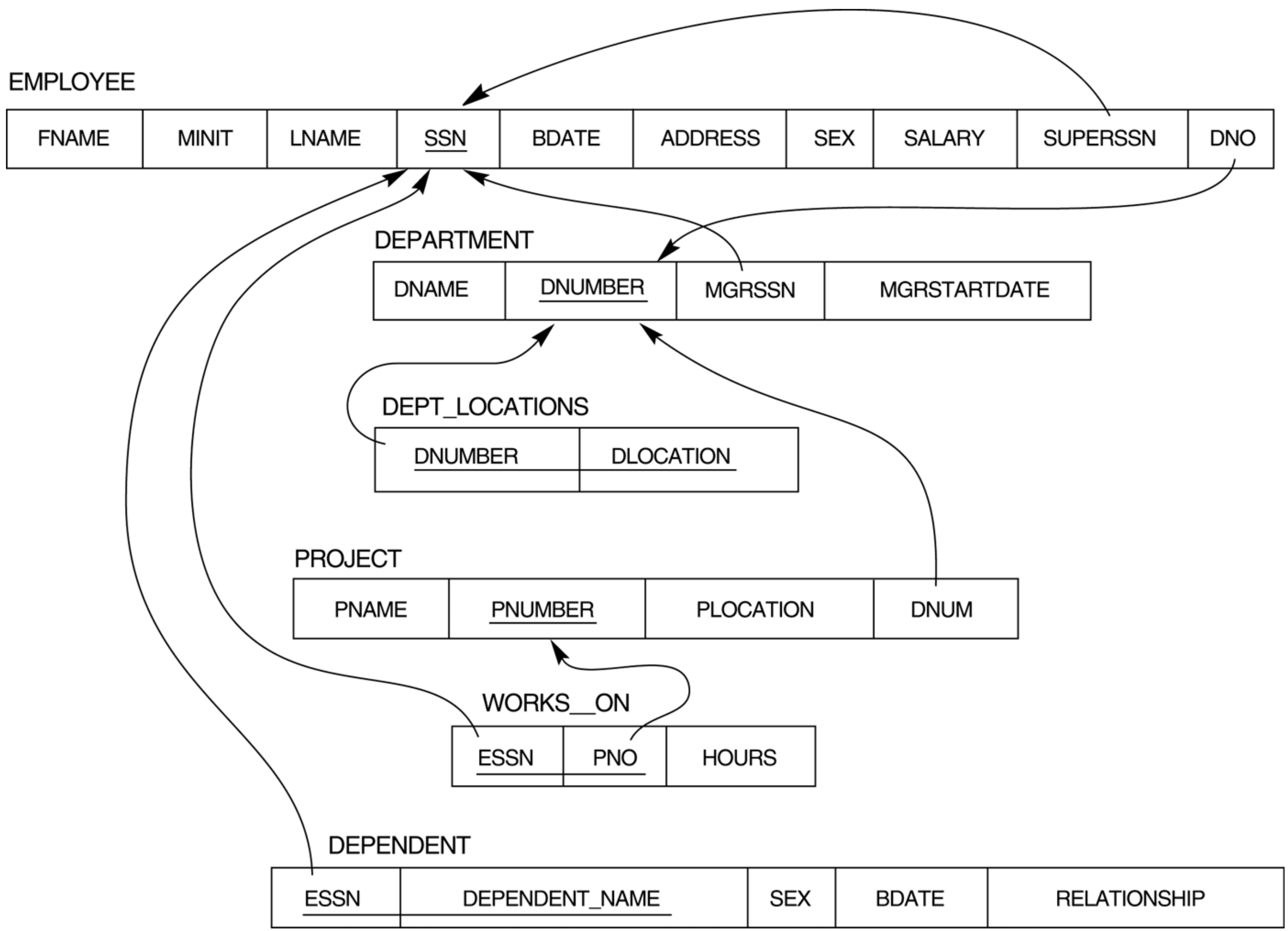
f. Retrieve the names of all employees who do not work on any project.

g. For each department, retrieve the department name and the average salary of all employees working in that department.

h. Retrieve the average salary of all female employees.

i. Find the names and addresses of all employees who work on at least one project located in Houston but whose department has no location in Houston.

j. List the last names of all department managers who have no dependents.


EMPLOYEE
FNAME
MINIT
LNAME
SSN
BDATE
ADDRESS
SEX
SALARY
SUPERSSN
DNO
DEPARTMENT
DNAME
DNUMBER
MGRSSN
MGRSTARTDATE
DEPT_LOCATIONS
DNUMBER
DLOCATION
PROJECT
PNAME
PNUMBER
PLOCATION
DNUM
WORKS__ON
ESSN
PNO
HOURS
DEPENDENT
ESSN
DEPENDENT_NAME
SEX
BDATE
RELATIONSHIP

EMPLOYEE	FNAME	MINIT	LNAME	SSN	BDATE	ADDRESS	SEX	SALARY	SUPERSSN	DNO
	John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
	Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
	Alicia	J	Zelaya	999887777	1968-07-19	3321 Castle, Spring, TX	F	25000	987654321	4
	Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
	Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
	Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
	Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
	James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	null	1

DEPT_LOCATIONS	DNUMBER	DLOCATION
	1	Houston
	4	Stafford
	5	Bellaire
	5	Sugarland
	5	Houston

DEPARTMENT	DNAME	DNUMBER	MGRSSN	MGRSTARTDATE
	Research	5	333445555	1988-05-22
	Administration	4	987654321	1995-01-01
	Headquarters	1	888665555	1981-06-19

WORKS_ON	ESSN	PNO	HOURS
	123456789	1	32.5
	123456789	2	7.5
	666884444	3	40.0
	453453453	1	20.0
	453453453	2	20.0
	333445555	2	10.0
	333445555	3	10.0
	333445555	10	10.0
	333445555	20	10.0
	999887777	30	30.0
	999887777	10	10.0
	987987987	10	35.0
	987987987	30	5.0
	987654321	30	20.0
	987654321	20	15.0
	888665555	20	null

PROJECT	PNAME	PNUMBER	PLOCATION	DNUM
	ProductX	1	Bellaire	5
	ProductY	2	Sugarland	5
	ProductZ	3	Houston	5
	Computerization	10	Stafford	4
	Reorganization	20	Houston	1
	Newbenefits	30	Stafford	4

DEPENDENT	ESSN	DEPENDENT_NAME	SEX	BDATE	RELATIONSHIP
	333445555	Alice	F	1986-04-05	DAUGHTER
	333445555	Theodore	M	1983-10-25	SON
	333445555	Joy	F	1958-05-03	SPOUSE
	987654321	Abner	M	1942-02-28	SPOUSE
	123456789	Michael	M	1988-01-04	SON
	123456789	Alice	F	1988-12-30	DAUGHTER
	123456789	Elizabeth	F	1967-05-05	SPOUSE

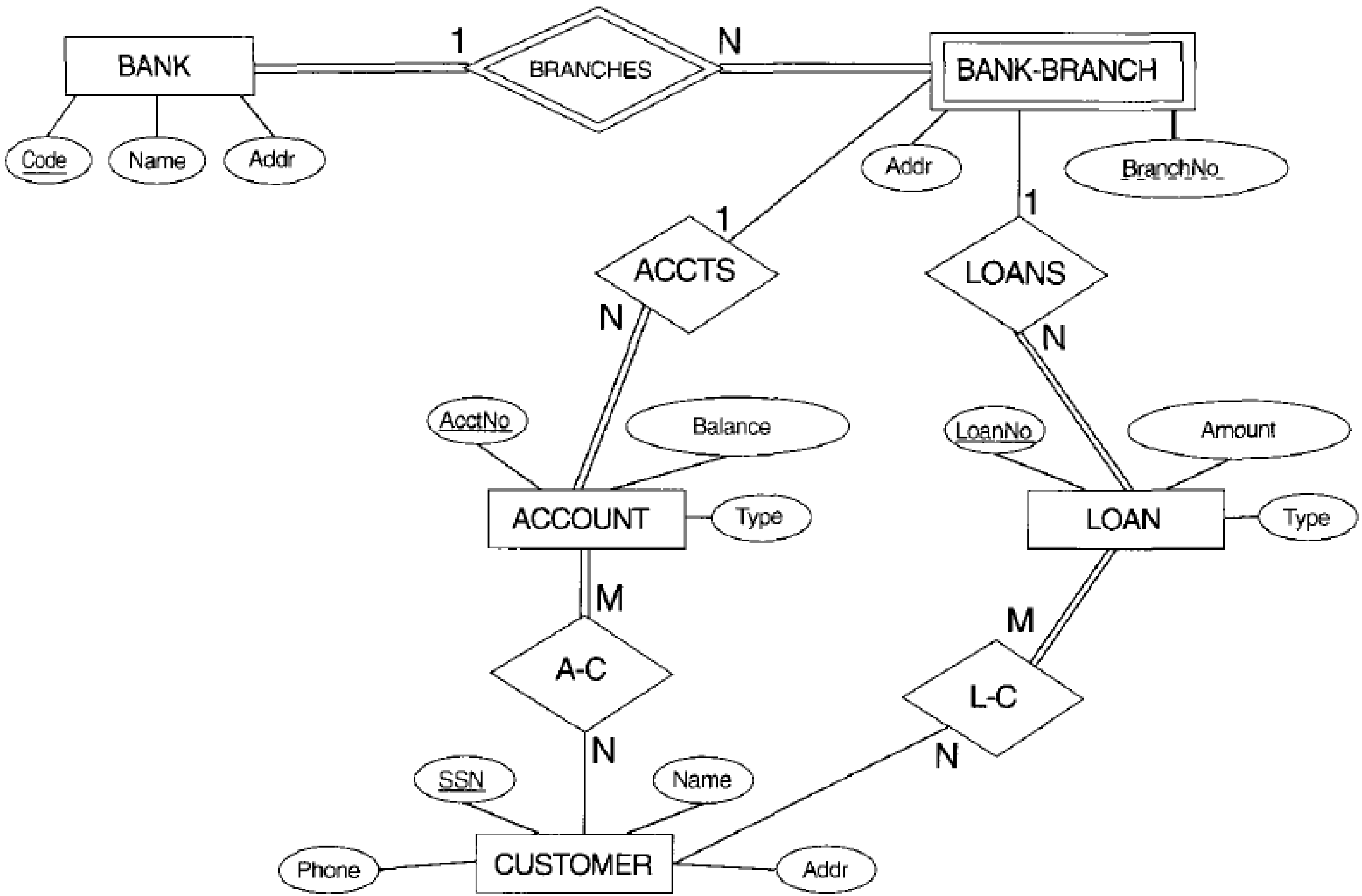


FIGURE 3.18 An ER diagram for a BANK database schema.

Mapping Exercise-2

An ER schema for a SHIP_TRACKING database.